

FRONT PANEL – 2 layers, 180 x 100 mm; copper = AGND, joined to GND only at the main-board star point NT1

The panel lies FLAT ON THE BACK of the main board (Decision #45). J1 / J2 / J3 are 2x20 FEMALE headers on the panel's INNER face (B.Cu); they mate J6 / J7 / J8, the male headers on the main board's bottom side.

Panel coordinates are mirrored in x against the main board (panel x = 180 – main x, panel y = main y), because the panel is drawn from its OUTER face. The main board's REAR edge is therefore the panel's TOP edge.

With that mirror, panel pad k mates main-board pin k on all three headers (the KiCad PinSocket footprint already carries the mating mirror); gen_panel.py proves it from the pad coordinates before the board is written.

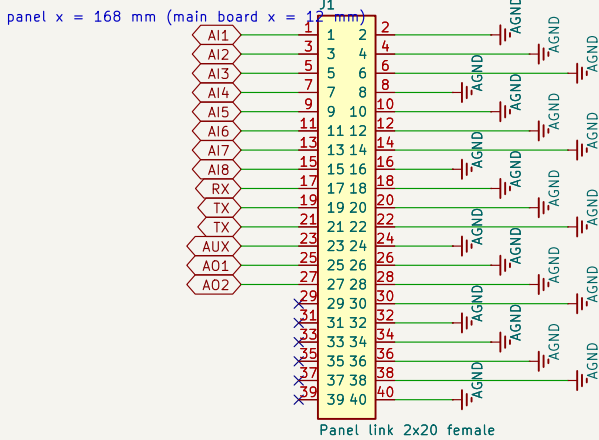
So the analog header J6 (main x = 12) is panel J1 at panel x = 168, the digital header J7 stays at x = 90, and the power header J8 (main x = 168) is panel J3 at panel x = 12.

Pins marked spare are not connected on the main board and are left unconnected here. SPARE (the 18th SMA position is empty; the fitted SPARE SMA) has its shield on AGND and its centre pin on TP1 only.

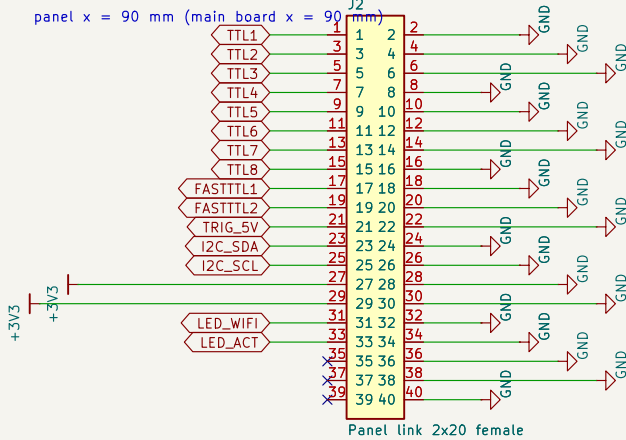
H5 / H6 are M2 (2.2 mm) holes for the OLED module; the shared library only has the M3 symbol, so their footprint is overridden to MountingHole:MountingHole_2.2mm_M2.

AO1/AO2: +−10 V, 49.9 ohm back-terminated. AI1..AI8: +−10 V, 1 kohm series on the main board. TRIG: 5 V into open circuit, about 2.4 V into 50 ohm. FAST1/FAST2 = the main-board nets FASTTTL1 / FASTTTL2.

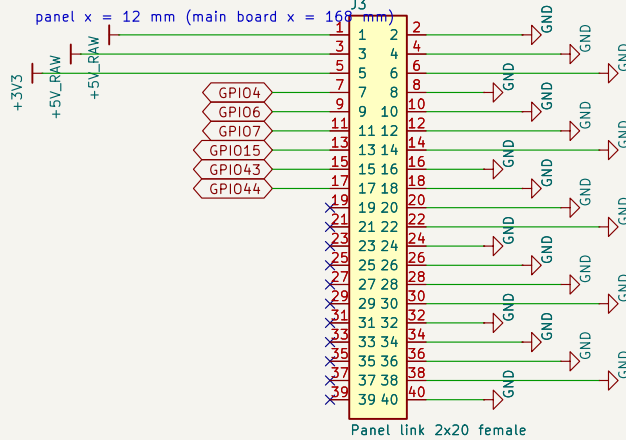
J1 – analog (inner face): mates J6, the analog header



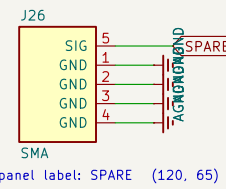
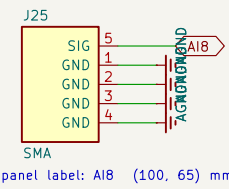
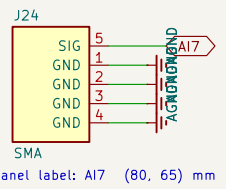
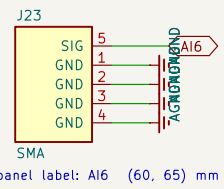
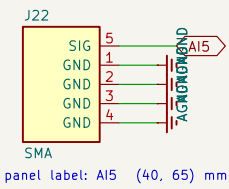
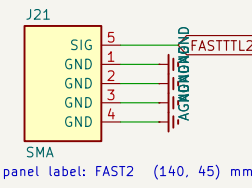
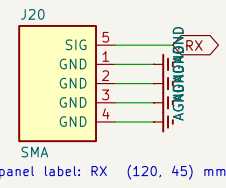
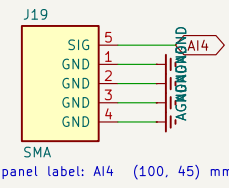
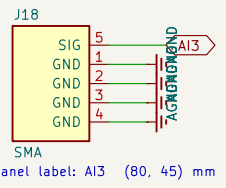
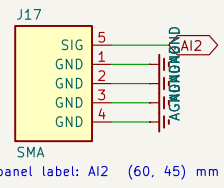
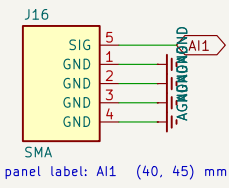
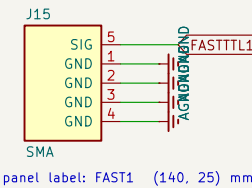
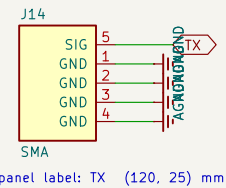
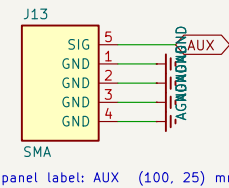
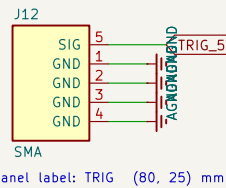
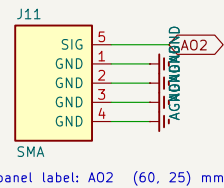
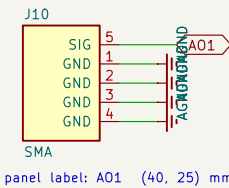
J2 – digital (inner face): mates J7, the digital header



J3 – power + spares (inner face): mates J8, the power + spares header



17 SMA on a 20 mm grid (3 x 6, the 18th position is empty). Shields on AGND; the panel copper is AGND on both layers.



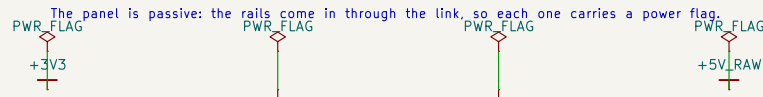
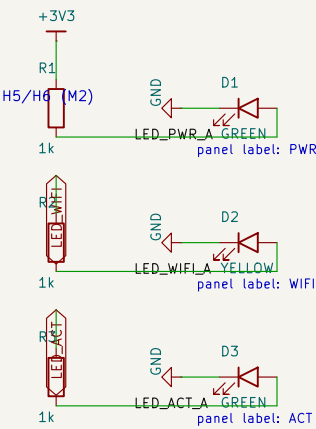
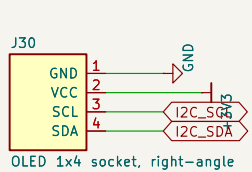
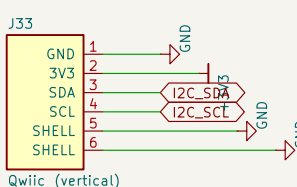
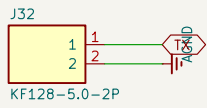
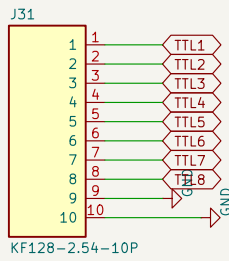
TTL1..8 out (5 V, 1 kohm series on the main board) + 2 x GND

TX coil terminal, beside the TX SMA (kept: user 2026–09–16)

Qwiic (vertical): sensor from the front; shell pads on GND

0.96 in I2C OLED (0x3C) on a right-angle socket: the module lies flat, held by H5/H6 (M2)

SPARE SMA: shield on AGND, centre pin on TP1 only (no link pin)



H1–H4: M3, same pattern as the main board (stand-offs). H5/H6: M2 for the OLED module.